Sustainable Development Goals

FUPOL and E-government

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The sustainable development goals commissioned by the UN in 2015 aim to end all forms of poverty, fight inequality and take measures to alleviate climate change by 2030. Goal 11 focuses on the development of sustainable cities and communities which is essential in a world where 60% of its inhabitants will live in urbanized areas by 2030 [4]. Goal 11 specifies the provision of safe housing, sustainable waste disposal and the protection of natural and cultural heritage however, this essay focuses primarily on the achievement of target 11.9*: "Enhance inclusive and sustainable urbanization and capacity for participatory, integrated and sustainable human settlement planning and management in all countries".* SMART cities transform the archetypal into a new, people centric city. Leaders of these cities make decisions transparently based on active participation of its citizens. The future policy modelling project (FUPOL) introduces a new governance and management system incorporating open policy design and innovative technology.

FUPOL is a research project funded by the EU currently partnered with over 15 countries investigating the integration of new technologies and methods for e-governance, e-policy and participation modelling[5]. FUPOL's approach to e-governance looks for the participation of all stakeholders and uses technology to make the process as transparent as possible. The platform builds on recent innovations such as crowd sourcing, multi-channel social computing and simulation. FUPOL specifies a complete policy life cycle model which integrates technology enabled community feedback in every step.

FUPOL was implemented in Zagreb, Croatia in 2013 for designing new policies centred on social infrastructure improvement and building a new autism centre with sensory park. FUPOL was used as a method of collecting opinions of a large group of stakeholders. The topics were launched on Zagreb city's social media pages, the Zagreb city website and blog using the FUPOL platform allowing citizens to share their thoughts on the proposed ideas. This created an opinion map capable of being geo-referenced. Additionally, the platform's feature allowing for social network aggregation was used in hot topic sensing permitting the extraction of the most hotly debated ideas. The feedback was integrated with FUPOL simulation technology which forecasts various scenarios based on mathematical modelling at evaluation workshops held for government administrators allowing them to choose the most suitable path for development. Civil servants were trained, given access rights and provided with continuous support in using the FUPOL platform. This ICT solution was successful in involving members of the community in all aspects of policy making and has over 350 likes on its Facebook page.

A limit to this ICT solution is the quality of data; FUPOL uses a single common data model for all sources whilst supporting data improvement by flagging or self-fixing. However, an issue with this is the recognition of weak data such as subjective values expressed over social media sites. The social media strategy is limited as many politicians are fearful of it as social media sites cannot be controlled. FUPOL are therefore developing a better Social Network Analysis platform which allows for the large variety of opinions expressed. FUPOL is not just a technological invention but a reformation of a political system and is therefore dependent on the responses of existing power holders within governments. Many city

administrators are fearful of change as they foresee PR nightmares emerging because of mis-communication between civil servants and citizens. The trepidation over reformation limits the speed and success of the FUPOL system uptake.

Another limit to the scale of this solution is the achievability of e-government in developing countries. This motivated the implementation of a FUPOL system in Mtwapa, Kenya with the partnership of the UN. Mtwapa has a population of roughly 50,000 and most of its inhabitants are living in slum dwellings. The FUPOL system sought the involvement of the community in slum upgrading management. An initial evaluation of the area saw that there is high mobile penetration of 74% as well as high speed internet available in nearly all the country making Mtwapa a suitable test environment for FUPOL. The technical environment for FUPOL was established as a Facebook page and blog where residents could suggest solutions and problems as well as keep track of recent development initiatives. Short text messages and an online learning system are other examples with much information collected from social media and the local newspapers.. The community was very positive towards the system as they felt their opinions were being heard: "e-governance makes it easier to consolidate people's thoughts to bring about solutions" assistant community mobiliser, Mtwapa [7]. Furthermore, FUPOL has been shown to succeed in a multilingual environment from the extraction of several hot topics from mixed language responses.

One limit to this ICT solution is the need to have ICT literate citizens. In Kenya FUPOL initiated a Community Mobiliser scheme which enabled non-ICT literate citizens to share their opinions. The success of this case study lies in the high mobile penetration in Kenya which highlights a weakness of e-participation as it cannot be successfully implemented in lesser developed countries with limited technology. Another limit to the scheme is the management of the platform which requires the hiring of trained staff dependent on budget allowances. One weakness of the FUPOL scheme is time efficiency, it takes a long time for the scheme to be instantiated in a developing country thus requiring both the government and the communities to be well prepared.

The economic costs of the implementation of a FUPOL system are dependent on the development, implementation and maintenance of the project. It is difficult to calculate the long-term costs of instantiating an e-government system leading to a focus on short-term costs which restricts the expansion of e-government and results in questions over its sustainability. Additionally, the success of a FUPOL system relies on universal access to the internet. In places where this is not already achieved there is a cost in providing society with the technology. One risk associated with instantiating a FUPOL system is outsourcing policy making. The dependence on a technology based system may not translate well to countries with a complicated government network. The FUPOL pilots have already created IPL conflicts where numerous parties are merged in the policy making process and further legislation needs to be developed to rectify the questions over data ownership.

The introduction of FUPOL and other e-government schemes can have repercussive effects on other sustainable goals. One of these is goal 10: *"Reduce Inequalities*". FUPOL works to provide better governments by involving citizens in policy making, this in turn reduces inequality as impoverished communities can effect change. However, FUPOL may increase inequality as the political reform in some areas of a country may lead to gentrification forcing poorer citizens out of the area. The implementation of e-governments will work towards goal 16: *"Reducing Corruption"* as the policy making process is made increasingly transparent to its citizens the opportunities for governmental corruption will decrease. Additionally, this ICT solution works towards the achievement of goal 4: "*Promoting life-long*

education" as the FUPOL system aims to educate citizens in local politics and the use of IT. E-government systems also have widespread benefits as they are integrated with other sustainable development projects such as the FUPOL implementation in Kenya, used alongside slum upgrading which is working to achieve goal 1: *"No Poverty".*

It is not only cities being made more intelligent, technology is being applied across all sectors even in the agricultural industry. Goal 2 of the sustainable development goals is the aim to achieve no hunger by 2030. The agricultural sector is going to face significant challenges in addressing this as to feed 9.6 billion people food production must increase by 70%[9]. This is tested by the lack of available land and fresh water needed to grow crops as well as the unpredictable factor of climate change. One ICT solution conceived to address these issues is SMART floating farms. Still in the development stage, early prototypes promise year-round, large scale food production with the possibility of creating emergent economies.

The farms are designed to float on large bodies of water; using a multi layered design they have designated sectors to produce fish, crops and solar energy. The multi layered strategy allows for higher yield per metre squared than land based farms. The floating farms also employ sustainable water management and waste disposal and their locations will be at centres of consumption which reduces the carbon emission of consumers as well as providing job opportunities to the locality. This ICT solution has massive potential as it offers solutions to multiple sustainable goals as well as overcoming the fundamental issue of land scarcity to overcome world hunger.

The scale of this solution is limited by open bodies of water located in proximity to large urban developments; dry and barren countries that are landlocked will not benefit from this solution. Additionally, the farms will not be able to be implemented in environments currently protected as natural heritage and consideration will be paid to existing water based habitats. Politically this solution will be tested by the strong hold supermarkets have over the existing economy. The floating farms are not designed to grow every product currently on the market which makes them less desirable than current supermarkets.

To encourage uptake, the government could instantiate tax subsidies to companies or households who use the floating farms as well as increasing penalties for high carbon emissions. The farms need to be designed to be visually stunning thus creating incentive for people to visit them over supermarkets. A limiting factor in the uptake of floating farms is the high initial cost, however, as zero hunger is the UN's top priority the initial investment will be fronted providing the farms are self-sufficient and sustainable. The promise of an emergent economy will also encourage private investors.

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